

EXAM 7 – SPRING 2014

27. (2 points)

a. (1 point)

Describe one benefit and one drawback of using a probability density function to model a company's capital when building an internal model for Solvency II.

b. (1 point)

Discuss one alternative to using a probability density function to model a company's capital when building an internal model for Solvency II and describe one advantage of this approach.

END OF EXAMINATION

SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 27	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE: C8: Evaluate best practices in risk measurement, modeling, and management of various financial and non-financial risks faced by an entity.
SAMPLE ANSWERS	
Part a: 1 point	
<u>Benefit</u>	
<i>Sample Answer 1</i>	
By calculating a few points of the distribution, you can fit a curve to find other values.	
<i>Sample Answer 2</i>	
It is easy to fit a pdf to a distribution of capital, even if only a few points in the distribution are available. The pdf can be used to smooth out the distribution and calculate interpolated or extrapolated quartiles, confidence intervals, etc.	
<u>Drawback</u>	
<i>Sample Answer 1</i>	
Does a poor job of representing the full range of possible real-world outcomes	
<i>Sample Answer 2</i>	
The selected probability density function may under fit the tail of the distribution which is the most important part for solvency purposes.	
<i>Sample Answer 3</i>	
It has difficulties in handling steps in the distribution, such as when reinsurance contracts are applied.	
Part b: 1 point	
<i>Sample Answer 1</i>	
Use the sampled capital values (from the individual risk capital distribution) to represent the distribution. A copula can then be used to combine the sampled capital values. Advantage: This avoids the need to approximate the distribution.	
<i>Sample Answer 2</i>	
A better approach would be to use a model that allows risks to vary stochastically. Advantage: This would allow for non-linear effects between risks and of the impact of risks in the capital required.	

SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 3

You can use a bootstrap procedure to essentially come up with your own distribution based on your observed losses.

Advantage: you would no longer be tied to picking a known pdf for your losses.

EXAMINER'S REPORT

Candidates generally performed poorly on this question. There appears to be a large number of candidates confused by probability density functions, probability distribution functions, and cumulative distribution functions. Candidates are assumed to know the differences and relationships between these functions.

Part a

Candidates struggled to provide a benefit of using a probability density function for modeling capital. Candidates performed better when articulating the drawbacks of using density functions. A variety of responses received partial credit.

Part b

Candidates generally performed poorly on this part; a substantial number of candidates did not provide a true alternative to probability density functions. Instead, they provided methods that were improvements or modifications to density functions. These adjusted PDF methods received partial credit. Some candidates provided alternatives that were density functions without any modification and the difference was in name only; these responses received no credit. Some candidates lost points when they provided an alternative without describing it adequately.